

CommissionLens

Commission-Adjusted Alpha Prediction in Mutual Funds

Finance & Economics Club - DIY Project 04

Problem Statement

1. India has over 90 million demat accounts, but many retail investors ignore the critical expense ratio gap between regular and direct mutual fund plans.
2. Regular plans charge an additional 0.5% - 1.5% annually as distributor commission.
3. Due to compounding, this small difference can erode Rs. 8 - Rs. 12 lakh from a Rs. 5,000 monthly SIP over 20 years.
4. The higher cost is ONLY justified if the fund generates sufficient excess return (alpha) over the benchmark to offset the commission.
5. Currently, no systematic ML framework exists to predict whether a fund will continue to justify this fee. CommissionLens solves this.

Goals & Methodology

- > Data Collection: Synthesized/ingested 5 years of quarterly data across 250 Indian Equity Mutual Funds, including macroeconomic constraints (Repo Rate, CPI, FII/DII).
- > Feature Engineering: Computed rolling 4Q/8Q Sharpe ratios, trailing volatility, information ratios, beta, and tracking error.
- > Target Definition: Binary classification labelling fund-quarters as 'Justified' if Next Quarter Gross Alpha > Expense Gap.
- > Temporal Splitting: Strict train/val/test split across time to prevent look-ahead bias in financial markets.

ML Models & Performance

Models Trained:

- Regression (Predicting Net Alpha): Linear Regression, Random Forest, XGBoost
- Classification (Predicting Justification): Logistic Regression, Random Forest, XGBoost

Evaluation Highlights (Test Set):

- Best Regression Model: Random Forest (RMSE ~0.028)
- Best Classification Model: Random Forest
- We successfully evaluated Accuracy, Precision, Recall, F1 Score, and Precision at Top Decile.

Conclusion: Tree-based models effectively captured the non-linear relationship between macroeconomic shifts, fund turnover, and alpha generation.

SHAP Explainability

We applied SHAP (SHapley Additive exPlanations) TreeExplainer to demystify our Random Forest model and identify the top drivers of commission justification.

Top predictive features discovered:

1. Rolling Volatility (8Q & 4Q)
2. Rolling 4Q Return
3. Log AUM
4. Beta (8Q)

Interpretation: High rolling returns and optimal volatility clustering strongly indicate whether a manager is actively providing value or just tracking the index.

SIP Back-Validation & Deliverables

Backtest Simulation:

- We simulated a monthly Rs. 5,000 SIP (Rs. 15,000 quarterly) over the test period.
- Compared three strategies: Naive Regular Plan, Model-Guided Regular Plan, and Direct Plan Benchmark.
- Outcome: The Model-Guided strategy effectively filtered out underperforming funds, minimizing dead-weight commission costs.

Final Deliverables Provided:

1. GitHub Repository with modular OOP codebase.
2. Jupyter Notebook research report.
3. Live Streamlit Interactive Dashboard.
4. Exported performance reports & SHAP dependency plots.